

Arrow Air Flight 1285R

Flight model: Douglas DC-8
Date: December 12, 1985

From: Gander, Canada
To: Campbell, US

Arrow air is a carter plane. There is a last minute runway change. The weather outside was snowy. The plane was carrying US military personals.

Problem

Seconds after take off, They experience a sever turbulence. The pilots were not able to bring the air speed to required levels. Then, the plane crashes into a dense tree cover area.

Investigation

The ground staffs have reported seeing ice build up in the wings of the windows of the plane. The investigators doubt if the crew had forgotten to de-ice the plane. The FDR found in the plane was old which means it could record only 4 parameter which includes altitude, speed etc in a aluminum foil, where as the modern FDR can record 100's of data in the modern storage system. It is found that the plane had flown in an extreme angle which made is not able to gain the required lift. The DVR didn't function properly and did not catch any sound in the cockpit. The FDR show after attaining the required speed for take off the plane took off then the speed started to decrease. The plane also seems to have taken off little longer than usual, which again points out to the accumulation of ice over the wings which would have reduced the acceleration. The weight calculation made by the pilots were normal passengers average which include adults and children but, this plane was carrying only adults (Military personal) which made the plane weight calculation lot less than actual. It was found that wrong weight calculation along with ice built up made the plane to crash. Also, the pilots have seem to have pulled up the nose just movements before crash which would increase the nose and drag which would reduce the speed and thrust further. It was found that even a ice cover as small as sandpaper would have impact on the aerodynamics. When both the overweight and icing together had made the plane to crash if one had been removed it would have saved the plane from crash.

Changes that were brought after the incident

The board advised on the importance of de-icing of plane in snowy conditions

Reference

1. The Deadly Takeoff at Gander — Mayday: Air Disaster
<https://www.youtube.com/watch?v=0UqS5rmLEuk>